

SIMATS ENGINEERING

ASSESSMENT TOOL 2

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning

COURSE OUTCOME COVERED

CO5: *Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)*

Assessment Tool Weightage: 35%

Dr G .A. SENTHIL

QUESTIONS: 10

Total Marks: 50

Annexure A

Scenario-Based Assessment

DURATION: 1.5 Hrs

Code of Conduct:

I Vignesh S / 192325002 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Vignesh S

SCENARIO-BASED ASSESSMENT — ANSWER

Question Chosen (Q4): A smart home system aims to reduce electricity consumption by automatically controlling lights, fans, and air conditioners based on user presence and weather conditions. Sketch a Reinforcement Learning model for this scenario by defining state space, action space, reward structure, and a suitable learning algorithm. (5 Marks)

1. Objective and Suitability of RL

The smart home must decide, at every moment, how to run its lights, fans and air-conditioners so that electricity use is minimised without sacrificing occupant comfort. These decisions are sequential and interact over time — pre-cooling now affects comfort and cost later — and they must be made under uncertainty about occupancy and weather. This is precisely the setting reinforcement learning is designed for: a controller (agent) repeatedly senses the home, takes control actions, and learns from the resulting comfort and energy outcomes to maximise long-term reward.

2. RL Formulation (Markov Decision Process)

The home is modelled as a Markov Decision Process. At each control step the agent observes the state, selects an action (appliance settings), the home transitions to a new state under its thermal dynamics and the occupancy/weather schedule, and returns a reward. The agent learns a policy π that maps states to energy-efficient yet comfortable settings.

Reinforcement Learning Model for a Smart Home Energy System

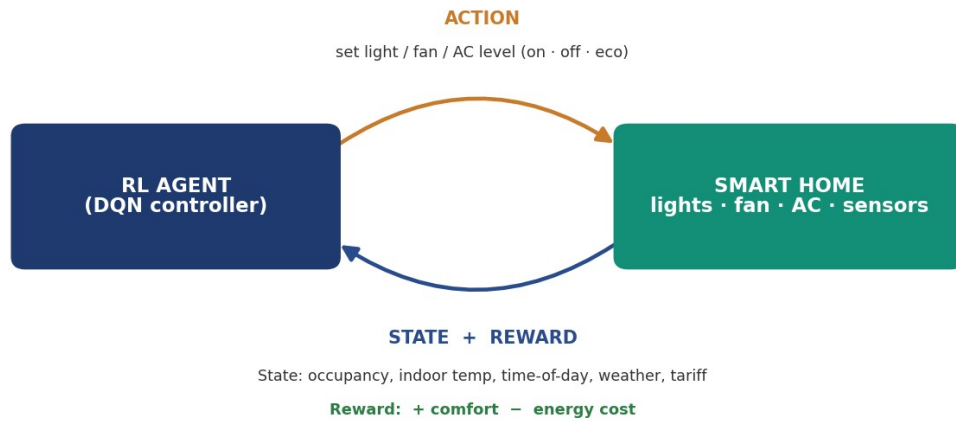


Figure 1: The agent–environment interaction loop for the smart-home energy controller.

3. State Space, Action Space and Reward Structure

Component	Definition in the smart-home MDP
State (S)	Occupancy (present / absent), indoor temperature (or deviation from set-point), time-of-day, outdoor weather (temperature, light level), and the current electricity tariff.
Action (A)	A discrete control level for each device — lights {OFF, DIM, ON}, fan {OFF, LOW, HIGH}, AC {OFF, ECO, COOL}. The joint setting is the action taken each step.
Reward (R)	+ comfort term ($- \text{indoor_temp} - \text{set-point} $ and adequate lighting, counted only when occupied) $- \lambda \cdot \text{energy cost (kWh} \times \text{tariff)}$. A large penalty is added if comfort or safety limits are violated.
Transition	Next state follows from the occupancy schedule, weather forecast, and the home's thermal response to the chosen action.
Discount γ	≈ 0.95 , so the agent values long-term daily savings rather than only the next step.

Reward design insight: the single scalar reward encodes two competing goals. The weight λ trades comfort against savings — a larger λ makes the controller thriftier, a smaller λ favours comfort. Counting the comfort term only when the home is occupied is what allows the agent to aggressively set appliances back while nobody is home.

4. Suitable Learning Algorithm

Because the state has several dimensions but the action set is discrete and modest, a Deep Q-Network (DQN) is a suitable choice. A neural network approximates the action-value function $Q(s, a)$; experience replay and a target network provide stable, sample-efficient off-policy learning from logged home data, which is valuable because a home produces only a limited amount of experience each day. If the state is coarsely discretised, plain tabular Q-learning also

works; if the actions become continuous (for example a continuously variable thermostat set-point), a policy-gradient / actor–critic method such as PPO or DDPG would be preferred.

5. Worked Example — One Day

Overnight the home is empty, so the energy term dominates and the agent keeps the AC and lights OFF. At 7 a.m. the occupant arrives to a cold room; the comfort term now applies, so the agent switches the AC to COOL/ECO to reach the set-point and turns lights ON. Through the mid-day empty hours it sets everything back again. In the occupied evening it conditions the home and lights the rooms. The chart below shows the learned policy tracking the baseline during occupied hours (shaded) but cutting consumption sharply when the home is empty.

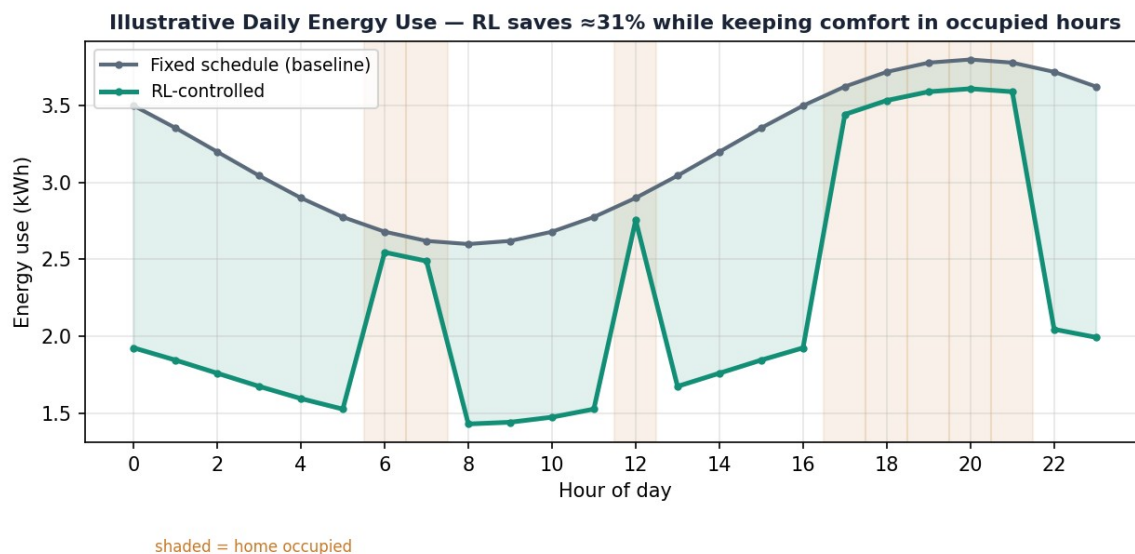


Figure 2: Illustrative daily energy profile — the RL policy saves about 31% overall while preserving comfort during occupied hours.

6. Practical Considerations

Three issues matter in deployment. First, the comfort–energy balance is sensitive to λ and should be tuned to the occupant's preference. Second, homes are non-stationary — seasons and habits change — so the policy must keep learning online. Third, exploration must be safe: the controller should never let temperature reach unhealthy extremes, which is best enforced with hard safety constraints layered on top of the learned policy, and comfort should be measured from occupant feedback or a reliable proxy rather than assumed.

7. Conclusion

The scenario maps cleanly onto an MDP with an occupancy- and weather-aware state, per-device discrete actions, and a comfort-minus-energy reward, learned effectively with a DQN. Such a model can cut household electricity use substantially — around 31% in the illustration — while keeping the home comfortable whenever it is occupied, and it adapts as usage patterns change.

RUBRICS FOR CALCULATION

Criteria	Weightage (%)	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Scenario	20%	Demonstrates clear and	Shows good understandin	Partial understanding	Misinterpret s or fails to

		complete understanding of the scenario and context	g with minor gaps	; misses key aspects	understand the scenario
Identification of Key Issues	20%	Accurately identifies all relevant issues and factors involved	Identifies most key issues with minor omissions	Identifies limited issues; some irrelevant points	Unable to identify key issues
Application of Concepts	25%	Applies appropriate concepts effectively to address the scenario	Applies concepts with minor errors	Limited or partially correct application	Unable to apply relevant concepts
Decision Making	20%	Makes well-reasoned, logical decisions with strong rationale	Makes reasonable decisions with some justification	Makes weak decisions with limited justification	No clear decisions made or rationale provided
Clarity of Response	15%	Response is clear, structured, and logically presented	Generally clear with minor issues	Somewhat unclear or poorly structured	Disorganized and difficult to understand